Chems Eddine Adel KEBAILY

Master's Student (M1) in Signal and Image Processing | AI Engineering | AI Agents & Generative AI

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Profile

Master's student in Signal and Image Processing at the Université du Littoral Côte d'Opale, moving into **AI engineering**: **AI agents**, **generative AI** and **prompt engineering**. Built three **AI agents** (administrative documents, CK-AI, a Telegram-to-Notion bot) with **LLMs**, OCR and n8n. Two internships at **Ooredoo Algeria** in web (Spring Boot, React, SQL/Oracle) and mobile (Flutter) development, following Agile practices. **Internship from 1 April 2027 (up to 5 months)**: AI agents, generative AI, ML/DL, vision, NLP.

Education

2026 – 2027 **Master's degree (M1) in Signal and Image Processing** (ongoing), Université du Littoral Côte d'Opale (ULCO), France

2022 – 2025 **Bachelor's degree in Computer Systems**, M'Hamed Bougara University of Boumerdès, Algeria

2022 **Baccalaureate in Technical Mathematics, Civil Engineering** (with honours), Ali Chekiri High School, Chéraga

Professional Experience

Mobile Development Intern — Ooredoo Algeria **Dec. 2024 – Jan. 2025**

- Cross-platform mobile apps with **Flutter** across the full **Agile** cycle; exposure to datacenter and **hybrid cloud** (GCP, on-premise).

Web Development Intern — Ooredoo Algeria **Dec. 2023 – Jan. 2024**

- Developed a web application for managing cash flows of point-of-sale (POS) terminals with **Spring Boot** and **React**.
- Analysed and processed data with **SQL** queries; maintained and optimised **Oracle** databases.
- Implemented functional enhancements, **unit tests**, technical documentation, code reviews and Git versioning.

Projects

AI Agent for Administrative Document Processing (French public services) **2026**

- **OpenCV** image preprocessing (denoising, **Sauvola** binarisation, deskewing), **OCR**, structured field extraction with an **LLM** (Claude API), completeness checks; **n8n** orchestration with **human-in-the-loop** validation, FastAPI, Docker.
- On 40 labelled documents: **97.6%** field accuracy, **100%** classification accuracy, OCR error reduced **90-fold** by preprocessing. Code

CK-AI & AgentGate — Claude API agent and agent evaluation (tool use, LLM judge) **2026**

- **CK-AI**: bilingual agent on the **Claude API** with **tool use** (**6 tools**: profile lookup, offer matching, 4 page actions), **guardrails** against prompt injection and abuse, **serverless** on Netlify. Demo
- **AgentGate**: tests agents on fixed cases, **traces** each call with its cost, scores answers (code checks, **LLM judge**), tests a **model router**, fails **CI** on a regression; **Haiku 5.5** matched **Fable 5.2** (12/12) at **1/70** of the cost. Code

DMS & Second Brain — Telegram → Notion bots (n8n, Gemini, OpenCV) **2026**

- **DMS**: note photos become searchable PDFs in Notion (**n8n**, Gemini, **FastAPI**, **OpenCV**, **OCR**); pages turned upright 12/12 (Tesseract: 0/12). Code
- **Second Brain**: text, voice or receipts become expenses, tasks and notes in Notion, plus a chat mode over my data (**LLM** router, **structured outputs**, **PostgreSQL**); 65/66 test cases correct. Code

Phishing Detection with Machine Learning — Bachelor's Final Project (research) **2025**

- Comparison of **16 classification models** (**XGBoost**, LightGBM, random forest, neural networks…) on 100,077 URLs with **Python** and **scikit-learn**; best: XGBoost, **89.6%** accuracy, +10.6 points over hand-written rules. Report

Skills

Generative AI & agents	AI agents, prompt engineering, Claude and Gemini APIs, structured outputs, n8n, FastAPI
Machine learning & vision	PyTorch, TensorFlow, scikit-learn, Vision Transformer, transfer learning, OpenCV, OCR
Programming	Python, C, C++, Java, JavaScript, PHP, SQL (Oracle), HTML/CSS
Development	Spring Boot, React, Node.js, Flutter, REST APIs
Tools & methods	Docker, PostgreSQL, Git/GitHub, Linux, Windows Server; Agile, code reviews, unit testing
Languages	Arabic (native), French (B2), English (B2)

Student Activities and Training

2024 – 2026 **DevUp Club, University of Boumerdès** — Visual Department co-manager, Development member

Feb. 2025 **"AI to Fullstack" Bootcamp** — DevUp Club (grade: Very Good)